

eSpark Consulting Group

Project Proposal

MVP Ad Contest SaaS Platform

1. Project Overview

This proposal outlines the development plan for a web-based SaaS platform where advertisers can launch 3-day ad contests across major platforms such as TikTok, Meta, Google Ads, and YouTube. Creators will compete by submitting real ads, and the platform will automatically track performance, identify the winning ad based on selected KPIs, and trigger payout upon contest conclusion.
The MVP will operate in an invite-only mode where interested users can join a waitlist. Upon approval, they will receive a unique signup link. This phased rollout allows for identifying pain points and continuously refining the platform prior to a full public launch.

The system is designed for three distinct user roles: **Advertiser, Creator,** and **Admin/Staff**.

2. Core Features

User Roles & Flows

Advertiser:

- Sign up and connect ad account
- Launch contest (choose platform, campaign type, KPI template)
- Pay listing fee and prize purse (via Stripe)
- View live leaderboard and scale winning ads
- Access billing and payout records

Creator:

- Sign up and link payout account (Stripe Express)
- Browse active contests and submit up to three creative concepts per contest
- Track live performance and earnings history

Admin/Staff:

- Create and manage KPI templates and platform guardrails
- Issue refunds, ban accounts, and review audit logs

MVP Feature Checklist

- **Authentication & RBAC:** Implemented using **Clerk.dev** with roles for advertiser, creator, and admin.
- **Contest Engine:** Supports KPI-driven campaigns with platform/campaign type cascades, 72-hour timers, finalist caps, and auto-lock on winner detection.
- **Metrics Pipeline:** Pulls ad performance stats every 30 minutes using **Windsor.ai** or native ad APIs, stored in **Supabase Postgres**, and visualized in real-time using **WebSockets**.
- **Payments: Stripe Connect Express** for escrow handling, prize purse deposits, and payout distribution with 60/40 revenue sharing (creator/platform), plus a 4% success fee on scaled spend.
- **Frontend:** Built with **React 18**, **Tailwind CSS**, and **Vite**; dedicated layouts for advertisers and creators.
- **Backend:** Developed using **Node.js** with **Fastify** (for performance), **async** operations via **async/await**, **Supabase Postgres** for data, and **Upstash Redis** for job scheduling.
- **Testing & DevOps: Jest** for backend testing, **Cypress** for front-end smoke tests, **ESLint** for linting; deployed via **Replit Deployments** with a container-based architecture.
- **Documentation:** Complete README for local setup and OpenAPI-based API documentation.

3. Milestones & Timeline Breakdown

Milestone 1: UI/UX Design

- **Description:** The project will begin with the design of all critical user flows using Figma. This will include screens for advertiser sign-up, contest creation, leaderboard viewing, creator onboarding, concept submission, and admin panels. Each screen will be designed for responsiveness and accessibility, with clear navigation and consistent visual language. Developer-ready annotations and components will be included to support a seamless handoff to engineering.
- **Timeline: 10 Days**

Milestone 2: Core Setup

- **Description:** This phase establishes the application's infrastructure. **Clerk.dev** will be used to manage authentication and role-based access control. **Stripe Connect Express** will be integrated to allow advertisers to deposit funds and creators to receive payouts. The **Contest Engine** will be implemented to allow advertisers to define campaigns with platforms, campaign types, KPIs, budgets, and time constraints. Business logic for finalist limits and automated prize handling will also be introduced in this phase.
- **Timeline: 3 Weeks**

Milestone 3: Metrics Integration & Leaderboard

- **Description:** Ad performance data will be fetched every 30 minutes via **Windsor.ai** or native APIs (TikTok, Meta, YouTube, or Google Ads). This data will be stored in a **Supabase Postgres** database and processed for contest evaluation. A **WebSocket**-based system will broadcast live contest results on a dynamic leaderboard. Automated winner detection based on the selected KPIs will be configured to trigger prize purse release upon contest completion.
- **Timeline: 3 Weeks**

Milestone 4: Admin Tools & Testing

- **Description**: An administrative dashboard will be developed to manage the platform. This includes tools for editing KPI templates, setting guardrails for campaign creation, viewing audit logs, and moderating user accounts. The testing suite will be implemented in parallel using **Jest** for backend logic and **Cypress** for UI integrity checks, ensuring functional reliability across the system.
- **Timeline: 2 Week**

Milestone 5: Final Polish & Deployment

- Description: This phase will focus on finalizing UI responsiveness, optimizing performance, and completing a comprehensive QA cycle. Minor bugs and inconsistencies will be addressed. The application will be deployed to a secure, TLS-enabled staging environment (Replit or Fly.io), seeded with demo data for final testing and stakeholder walkthroughs.
- **Timeline: 1 Week**

Milestone 6: Feedback Loop & Documentation

- Description: The final milestone is reserved for stakeholder feedback and final polish. Any edge cases, usability issues, or overlooked bugs will be resolved. Complete documentation will be delivered, including a local setup guide, developer usage notes, and OpenAPI-based API docs.
- **Timeline: 1 Week**

4. Technology Stack

Layer	Technology	Notes
Frontend	React 18, Tailwind CSS, Vite	Lightweight, modern, fast development cycle
Backend	Node.js, Fastify, async/await	Async-first backend framework, optimized for performance
Auth	Clerk.dev	Role-based auth for advertiser/ creator/admin
Database	Supabase Postgres	Row-level security, hosted Postgres
Real-Time	WebSocket	For live leaderboard updates
Metrics	Windsor.ai / Native APIs	For pulling ad performance data every 30 min
Payments	Stripe Connect Express	For escrow and creator payout management
DevOps	Replit Deployments	Simple container-based CI/CD
Testing	Jest, Cypress, ESLint	Automated test coverage and linting

5. Costing

Milestone / Phase	Description	Cost Estimate (USD)
Milestone 1: UI/UX Design	Figma screens for all critical user flows with annotations and developer assets	**$500**
Milestone 2: Core Setup	Auth (Clerk), payments (Stripe), contest engine, role-based access	**$2,000**
Milestone 3: Metrics Integration & Leaderboard	Windsor/native API integration, Supabase DB, WebSocket real-time leaderboard	**$2,000**
Milestone 4: Admin Tools & Testing	Admin panel tools, KPI manager, audit logs, Jest, Cypress	**$3,000**
Milestone 5: Final Polish & Deployment	UI polish, performance tuning, deployment (Replit/Fly.io)	**$1,000**
Milestone 6: Feedback Loop & Documentation	Stakeholder feedback handling, bug fixes, OpenAPI docs, local setup guide	**$800**
Total Fixed Cost	-	**$9,300**

- **Payment Terms:**

  A **40% upfront payment ($3,720)** is required before project kickoff to secure resources, begin design, and initiate core development. The remaining amount will be invoiced and distributed across milestone completions, with a detailed payment schedule to be agreed upon at contract signing.

6. Conclusion

This proposal outlines a clear, actionable roadmap to build a scalable and feature-complete MVP of the Ad Contest SaaS platform. After careful evaluation of the project requirements and performance considerations, we have decided to use Node.js instead of Python/FastAPI for the following reasons:

- **Performance & Scalability:** Node.js's asynchronous, non-blocking event-driven architecture makes it ideal for handling large volumes of concurrent connections, as seen in our real-time leaderboard and metrics synchronization. This ensures reduced latency and improved performance across the application.

- **Optimized for Real-Time Operations:** The ability to handle real-time updates (such as contest metrics every 30 minutes) with minimal resource consumption is a key advantage of Node.js. This ensures a smooth, low-latency user experience.

- **Better Handling of I/O Operations:** Node.js excels at I/O-bound tasks like fetching data from multiple ad APIs (TikTok, Meta, Google Ads) every 30 minutes. It can handle concurrent requests and interactions more efficiently than Python in such use cases.

- **Our Expertise:** Our team has extensive expertise in building high-performance, optimized platforms using Node.js. This allows us to leverage our knowledge to deliver a solution that's highly optimized and scalable.

- **Ecosystem & Community Support:** Node.js offers an extensive ecosystem of libraries and tools, as well as a vibrant development community. This significantly reduces development time, helping us to launch the MVP faster.

- With this technology stack and roadmap, we are confident in delivering a high-quality, robust platform within 3 months. We look forward to collaborating with you to bring this innovative platform to market.